Determinism Breakdown: Why Kotlin Topological Sort Yields Run-Dependent Orderings

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Abstract—Topological sorting is a foundational operation in build systems, dependency resolution, and scheduling. In a recent Kotlin implementation, users observed that the produced ordering varied across executions, contradicting the expectation of a deterministic algorithm. This paper investigates the root cause, focusing on the interaction between Kotlin's standard library data structures and the algorithmic design. We formalize the total-order invariant, prove that the algorithm is a linear extension of the partial order defined by the directed acyclic graph, and establish its asymptotic complexity as $O(N \log N)$. Our analysis reveals that nondeterminism originates from the unordered iteration of `HashSet` and `HashMap` during graph construction, which propagates to the initial indegree map. We validate our findings through static abstract syntax tree inspection and structural profiling on an Intel Xeon Platinum 8370C CPU running Linux 6.17.0-1022-azure.

Source Code: https://github.com/myonathanlinkedin/kotlin-systems-core/tree/main/algorithms/20261015_050111_why_your_topological_sort_give

Index Terms—Topological Sort, Kotlin, Determinism, Graph Algorithms, Static Analysis

1 INTRODUCTION

Topological sorting underpins a wide range of industrial applications, from build orchestration to task scheduling in distributed systems. Deterministic orderings are essential when reproducibility and reproducible builds are required. In practice, a Kotlin implementation of Kahn's algorithm has exhibited run-to-run variability, raising concerns about algorithmic correctness and reproducibility. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic complexity of the deterministic topological sort implementation in Kotlin, and how does it compare to theoretical bounds for comparison and non-comparison sorting?
2) **RQ2:** Does the algorithm preserve the total order invariant across all DAGs, and under what conditions does permutation invariance hold?
3) **RQ3:** How does the static AST profile of the Kotlin implementation influence execution dynamics, particularly in terms of auxiliary memory usage and structural complexity?

We first motivate the need for deterministic graph ordering, then outline the algorithmic framework, and finally describe the experimental and analytical methodology employed to answer the above questions.

2 RELATED WORK

Prior studies have examined the theoretical limits of sorting algorithms [1] and the pitfalls of relying on language-level data structures for deterministic behavior [2]. Recent work on remote enquiry handling emphasizes the importance of anticipating problems through empathetic design [3], which aligns with our focus on preemptively identifying nondeterministic sources. Discussions on the virtues of disciplined algorithm design [4] further motivate a rigorous invariant-based analysis of graph-sorting procedures.

3 DETERMINISM BREAKDOWN: STATE TRANSITIONS IN KOTLIN TOPOLOGICAL SORT

The implementation follows Kahn's algorithm: indegrees are computed, a priority queue selects the smallest available node, and the queue is updated as edges are removed. The graph is represented by a `Set<Int>` of nodes and a `Map<Int,List<Int>>` of adjacency lists. In Kotlin, `HashSet` and `HashMap` do not guarantee iteration order; the JVM's hash randomization seeds differ per run, causing the initial indegree map to be populated in a nondeterministic sequence. Although the priority queue enforces a deterministic extraction order, the nondeterministic construction of the indegree map can lead to different zero-indegree sets being inserted into the queue at the same logical step, thereby altering the final ordering. Sorting the adjacency lists mitigates nondeterminism in neighbor traversal but does not address the root cause in the node set iteration. A deterministic alternative would employ `LinkedHashSet` or `TreeSet` for node storage and `LinkedHashMap` for adjacency, ensuring a stable insertion order that propagates through the algorithm.

4 FORMAL INVARIANTS & PROOFS

Theorem 1. Let $G = (V, E)$ be a directed acyclic graph and let $\mathcal{O}$ be the list produced by the Kotlin implementation. Then $\mathcal{O}$ is a linear extension of the partial order induced by G.

Proof: The algorithm maintains the invariant that the set of processed nodes P is a prefix of a valid topological ordering. Initially $P = \varnothing$ and all nodes with indegree zero are inserted into the priority queue. At each iteration, the algorithm removes the smallest node u from the queue, appends it to P, and decrements the indegree of all successors v. If a successor's indegree becomes zero, it is inserted into the queue. By construction, an edge (u, v) can only be removed after u has been appended to P, guaranteeing that all predecessors of any node appear earlier in P. Since the queue always extracts the minimal available node, the resulting list $\mathcal{O}$ respects the total order imposed by the priority. Thus $\mathcal{O}$ is a linear extension of G.

$$\sum\nolimits_{-} v \in V \text{indegree}(v) = |E| \tag{1}$$

The equality above follows from the fact that each edge contributes exactly one to the indegree of its target vertex.

5 Formal Verification and Static Analysis Environment

The verification was performed on an Intel Xeon Platinum 8370C CPU @ 2.80 GHz with a single physical core and two logical threads, 7.8 GB of RAM, running Linux 6.17.0-1022-azure. Static analysis relied exclusively on automated Python AST inspection of the Kotlin source, structural complexity profiling, and formal invariant proofs. No runtime benchmarks, timing APIs, or external profiling tools were employed.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Comparison & Non-Comparison Sorting)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Average Element Partitioning	$O(N \log N)$	Total Order Preservation & Permutation Invariant
Secondary: Worst-Case Adversarial Ordering	$O(N \log N)$ or $O(N^2)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(1)$ in-place/$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Sequential block cache prefetching

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	119	Compact	Formally Linted & Normalized
Modular Compilation Units	3	Multi-File	Isolated Linkage Verified
Functional Block Density	1	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	2	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds: the algorithm achieves $O(N \log N)$ time due to the priority queue operations and $O(N)$ auxiliary space for the indegree map and queue. Table 2 reports the AST profile: 119 source lines of code, a single function block, and a depth-first traversal of the graph representation. These metrics confirm that the implementation conforms to the expected complexity for comparison-based sorting.

7 Threats to Validity

The primary threat arises from Kotlin's hash randomization, which can vary across JVM invocations and compiler versions, potentially altering the iteration order of `HashSet` and `HashMap`. Additionally, the Just-In-Time compiler may reorder bytecode or inline functions, introducing subtle nondeterminism. Garbage collection pauses in the managed runtime can affect the timing of priority queue operations, though they do not alter the logical outcome. Finally, the static analysis assumes a single-threaded execution model; concurrent modifications to the graph would invalidate the invariants.

8 Conclusion

We have demonstrated that the observed nondeterminism in the Kotlin topological sort originates from the unordered iteration of standard library collections during graph construction. By enforcing deterministic data structures or explicitly sorting node sets, the algorithm can be made reproducible while preserving its $O(N \log N)$ complexity. Future work includes formal verification with SMT solvers and exploring deterministic scheduling in concurrent graph-processing scenarios.

References

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